Journal of Molecular Catalysis B: Enzymatic 128 (2016) 65–72

Contents lists available at ScienceDirect

Journal of Molecular Catalysis B: Enzymatic

journal homepage: www.elsevier.com/locate/molcatb

# Effect of deep eutectic solvent mixtures on lipase activity and stability

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ARTICLE INFO

*Article history:*
Received 16 November 2015
Received in revised form 26 March 2016
Accepted 29 March 2016
Available online 30 March 2016

*Keywords:*
Deep eutectic solvents
Solvatochromic parameters
Lipase
Stability

ABSTRACT

DESs (deep eutectic solvents) have many potential applications as cosolvents or anhydrous reaction media for biocatalytic reactions, owing to their non-volatility, non-flammability, non-toxicity, biocompatibility, biodegradability, and low cost. In this work, choline chloride ([Ch]Cl)-based DESs and DES mixtures containing two hydrogen bond donors were used as cosolvents to enhance the activity and stability of *Candida rugosa* lipase in aqueous reactions. The activity of lipase in an aqueous solution of [Ch]Cl:urea:glycerol was 155% higher than that in buffer. The half-life time of lipase at 40 °C in an aqueous solution of [Ch]Cl:glycerol was enhanced by 9.2 times. The lipase showed the highest acid stability and base stability in the aqueous solutions of [Ch]Cl:glycerol:thiourea and [Ch]Cl:ethylene glycol:formamide, respectively. In general, glycerol-containing DES mixtures were very useful in enhancing the activity and stability of lipase, while formamide-containing DES mixtures could not efficiently enhance the activity and stability of lipase. To understand the effect of DES mixtures on the activity and stability of lipase in aqueous solution, four solvatochromic parameters of DES mixtures were determined. When the solvatochromic parameters of DES mixtures were correlated with the stability of lipase in aqueous solutions of DES mixtures, it was found that thermal stability and storage stability of lipase were associated with the hydrogen bond acidity of DES mixtures. Acid stability and base stability of lipase were correlated with polarity based on Reichardt's dye and the dipolarity/polarizability of DES mixtures, respectively.

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## 1. Introduction

Organic solvents have long been used as reaction media for non-aqueous enzyme reactions since the pioneering results of Klibanov [1]. Organic solvents as reaction media could greatly enhance the stability and selectivity of enzymes and could cause the shift of the chemical equilibrium. However nowadays, many studies are interested in the discovery of new reaction media for non-aqueous enzyme reactions because of the environmental toxicity, flammability, and volatility of organic solvents. From more than 10 years ago, many researchers to find alternative reaction media for enzyme reactions have focused on ionic liquids (ILs) [2,3]. ILs are organic salts that melt below 100 °C. Interest in ILs stems from their potential as "green solvents"; specifically, their non-volatile character and thermal stability make them attractive alternatives for volatile organic solvents [4,5]. It has been observed that the use of ILs enhances the reactivity, selectivity, and stability of enzymes [2–4,6,7]. Although ILs as reaction media have many advantages, the main limitations of ILs are their high cost, difficult synthesis, and purity requirement. In particular, the activity and stability of enzymes can be highly influenced by the impurities of ILs [8]. In addition, the green aspect of ILs is threatened nowadays because of their possible toxicity and low biodegradability [9].

Recently, to overcome the limitations of ILs, deep eutectic solvents (DESs) have been developed. DESs can be prepared simply by mixing an ammonium or phosphonium salt with a hydrogen bond donor (HBD), and purification of DESs is not required [10]. Choline chloride ([Ch]Cl) is the most frequently used cationic salt because it is biodegradable and costs less than US $960 per metric ton for 98% purity. Various HBDs such as alcohols, amides, amines, and sugars can be used to prepare DESs. The properties of DESs can be easily controlled by changing the mixing ratio of the cationic salt and HBD. The melting temperatures of some DESs are lower than room

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http://dx.doi.org/10.1016/j.molcatb.2016.03.012
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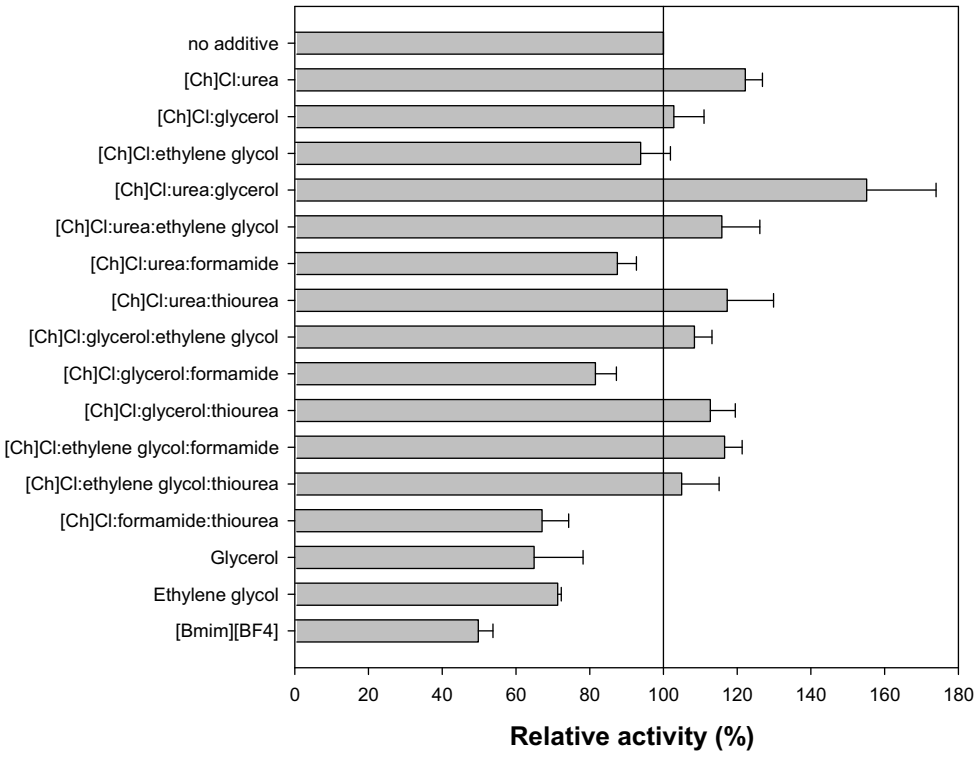



**Fig. 1.** Activity of lipase in aqueous solutions containing DES mixtures. The activity of lipase in buffer without additive was defined as 100%.

**Table 1**
Mixing ratios and water contents of the prepared DES mixtures.

| DES mixture | Molar ratio | Water content (%) |
|---|---|---|
| [Ch]Cl:urea | 1:2 | 1.65 |
| [Ch]Cl:glycerol | 1:2 | 0.91 |
| [Ch]Cl:ethylene glycol | 1:2 | 0.64 |
| [Ch]Cl:urea:glycerol | 1:1:1 | 0.59 |
| [Ch]Cl:urea:ethylene glycol | 1:1:1 | 0.63 |
| [Ch]Cl:urea:formamide | 1:1:1 | 0.61 |
| [Ch]Cl:urea:thiourea | 1:1:1 | 1.16 |
| [Ch]Cl:glycerol:ethylene glycol | 1:1:1 | 1.07 |
| [Ch]Cl:glycerol:formamide | 1:1:1 | 0.52 |
| [Ch]Cl:glycerol:thiourea | 1:1:1 | 1.16 |
| [Ch]Cl:ethylene glycol:formamide | 1:1:1 | 0.50 |
| [Ch]Cl:ethylene glycol:thiourea | 1:1:1 | 1.33 |
| [Ch]Cl:formamide:thiourea | 1:1:1 | 1.30 |

**Table 2**
Thermal stability of lipase in aqueous solutions containing DES mixtures.

| DES mixture | Half-life time of lipase at 40 °C (h) |
|---|---|
| Buffer (no additive) | 1.0 |
| [Ch]Cl:urea | 1.9 |
| [Ch]Cl:glycerol | 9.2 |
| [Ch]Cl:ethylene glycol | 5.3 |
| [Ch]Cl:urea:glycerol | 3.2 |
| [Ch]Cl:urea:ethylene glycol | 5.3 |
| [Ch]Cl:urea:formamide | 1.4 |
| [Ch]Cl:urea:thiourea | 2.0 |
| [Ch]Cl:glycerol:ethylene glycol | 8.9 |
| [Ch]Cl:glycerol:formamide | 5.0 |
| [Ch]Cl:glycerol:thiourea | 2.9 |
| [Ch]Cl:ethylene glycol:formamide | 6.1 |
| [Ch]Cl:ethylene glycol:thiourea | 6.0 |
| [Ch]Cl:formamide:thiourea | 0.9 |
| Glycerol | 5.3 |
| Ethylene glycol | 3.4 |

temperature. In general, DESs are cheap, easy to prepare, non-toxic, and biodegradable, while also providing the advantages of ILs such as non-volatility and high thermal stability. Therefore, the number of publications regarding the application of DESs is gradually increasing.

DESs as reaction media or cosolvents for biocatalysis have many more potential applications than organic solvents and ILs because of their non-toxicity, biocompatibility, biodegradability, and low cost. Recently, DESs have been used successfully as anhydrous reaction media for biocatalytic reactions. The activity, stability, and selectivity of enzymes such as lipase, protease, and epoxide hydrolase have been enhanced by using DESs as reaction media [9,11]. Gorke et al. studied lipase-catalyzed transesterification, aminolysis, and ring-opening polymerization in various DESs [12,13]. Zhao et al. showed that DESs based on cholinium salts and glycerol have merit as green solvents in lipase-catalyzed transesterification and preparation of biodiesel [14,15]. Durand et al. investigated the CALB-catalyzed alcoholysis of vinyl laurate [16].

Studies on the use of DESs as anhydrous reaction media for enzyme-catalyzed reactions have been gradually increasing, while very few results regarding the use of DESs as cosolvents for enzyme reactions have been reported. Gorke et al. reported that the conversion of styrene oxide catalyzed by epoxide hydrolase from *Agrobacterium radiobacter* was enhanced up to 20-fold in the presence of 25% [Ch]Cl:glycerol [12]. Lindburg et al. investigated the effect of DESs on the potato epoxide hydrolase-catalyzed hydrolysis of (1,2)-*trans*-2-methylstyrene oxide [17]. Huang et al. used cholinium-salt-based DESs as cosolvents for *Penicillium expansum* lipase (PEL) in aqueous solution [18]. The activity and stability of PEL could be increased up to 1.4-fold and 17.4-fold, respectively, by using [Ch]Ac:glycerol. Wu et al. used cholinium-salt-based DESs as cosolvents for horseradish peroxidase (HRP) in aqueous solution [19]. They reported that [Ch]Cl-based DESs have been found to be superior to the [Ch]Ac-based DESs in terms of promoting HRP activity. Although some DESs have proved to be good cosolvents for biocatalysis, the effect of DESs on the activity and stability of

enzymes should be studied more by examining the properties of DESs.

To help understand the solvent properties, solvatochromic parameters are very useful because they can be easily determined by measuring the shifts in UV–vis spectra for various dyes in solvents. The most widely used and measured solvatochromic parameters are $E_T^N$ and Kamlet-Taft parameters including dipolarity/polarizability ($\pi^*$), hydrogen bond acidity ($\alpha$), and hydrogen bond basicity ($\beta$). The solvatochromic parameters of ILs were recently used to help understand the equilibrium and kinetic phenomena including reaction rate, reaction selectivity, and solubility [20–25]. Solvatochromic parameters may also be very useful in helping to understand the solvent properties of DESs, but few results have been reported thus far. Therefore, solvatochromic parameters of DESs were measured in this study to help understand the solvent properties of DESs. The relationship between solvatochromic parameters of DESs and the enzyme action may explain the effect of DESs on enzymes.

In this work, various [Ch]Cl-based DES mixtures containing two HBDs were used as cosolvents for lipase-catalyzed hydrolytic reactions. When two HBDs for DESs are used, fine tuning of DES properties can be achieved and optimal DES mixtures as cosolvents can be easily found. In addition, solvatochromic parameters of DES mixtures were determined to help understand the solvent properties of DES mixtures. The direct comparison of solvatochromic parameters of DES mixtures with those of other organic solvents or ILs is very helpful for understanding the solvent properties of DES mixtures. Furthermore, the correlations between solvatochromic parameters of DES mixtures and the activity and stability of lipase in aqueous solutions of DES mixtures were analyzed to help understand the effect of DES mixtures on lipase.

## 2. Experimental

### 2.1. Materials

*N*,*N*-Diethyl-4-nitroaniline (**1**) was provided by Frinton Laboratories, Inc. (Hainesport, USA). Reichardt's betaine dye (**2**), 4-nitroaniline (**3**), choline chloride, urea, glycerol, ethylene glycol, formamide, thiourea, lipase from *Candida rugosa*, *p*-nitrophenyl palmitate, *p*-nitrophenol, isopropanol, Triton X-100, and acetonitrile were purchased from Sigma-Aldrich (St Louis, USA). All other chemicals used were of analytical grade and were used without further purification.

### 2.2. Preparation of DES mixtures

DES mixtures were prepared by mixing choline chloride with various HBD partners such as urea, glycerol, ethylene glycol, formamide, and thiourea (Table 1). The mixtures were stirred at 80 °C for 2 h and then cooled at room temperature. The water contents of the prepared DES mixtures were measured using a Karl Fischer titrator (870KF Titrino Plus, Metrohm).

### 2.3. Measurement of solvatochromic parameters of DES mixtures

A stock solution of 10 mM dye was prepared in methanol, and an appropriate amount of the dye stock solution was transferred to an Eppendorf tube. The methanol was removed by heating to 60 °C. DES mixtures (1 mL) were transferred to the Eppendorf tube with stirring at 25 °C. The absorbance was less than 0.8 and analyzed via a scan (probe **1**: 300–450 nm, **2**: 400–650 nm, **3**: 300–450 nm) of the UV–vis spectra using an Optizen 3220 UV spectrophotometer (Mecasys, South Korea). The solvatochromic parameters were calculated using the following equations:

$$E_T(30) = \frac{28591}{\lambda_{max}}, E_T^N = \frac{E_T - 30.7}{32.4} \quad (1)$$

$$\pi^* = 0.314 \times (27.52 - \nu(\mathbf{1})) \quad (2)$$

$$\alpha = 0.0649 \times E_T(30) - 2.03 - 0.72 \times \pi^* \quad (3)$$

$$\beta = 0.358 \times (31.10 - \nu(\mathbf{3})) - 1.125 \times \pi^* \quad (4)$$

where $\nu$ represents the wavelength of the longest wavelength absorption band of the probe in the region around 1000 $cm^{-1}$ and $\lambda_{max}$ [nm] represents the maximum wavelength due to the charge-transfer absorption band of the Reichardt's betaine dye (**2**).

### 2.4. Determination of lipase activity in aqueous solutions containing DES mixtures

The hydrolytic activity of lipase in aqueous solutions containing DES mixtures was determined by conducting a spectrophotometric assay. For the preparation of lipase solutions, 10 mg of lipase from *Candida rugosa* was added to 1 mL of 0.1 M phosphate buffer (pH 7.0) and shaken for 1 min. After centrifugation, the supernatant was diluted with phosphate buffer. The protein content was measured by a Micro BCA™ protein Assay Kit (Thermo Scientific, USA). The protein content of the lipase solution was 10 μg/mL. To 9.4 mL of 0.1 M phosphate buffer (pH 7.0) containing 10% DES mixture and 0.5% Triton X-100, 0.1 mL of lipase solution was added. Reaction was initiated by adding 0.5 mL of substrate solution prepared by dissolving 10 mM *p*-nitrophenyl palmitate in isopropanol and was sustained at 25 °C in a water bath with shaking at 120 rpm. Periodically (5, 10, 15, 25, and 30 min), 300 μL aliquots were removed, diluted with 300 μL of acetonitrile, and then centrifuged to obtain the supernatant. The activity was expressed as the initial rate and determined by measuring the increase in the absorbance at 400 nm caused by *p*-nitrophenol produced during the lipase-catalyzed hydrolysis of *p*-nitrophenyl palmitate. The content of *p*-nitrophenol produced by non-enzymatic reaction in each reaction media were subtracted as the background to calculate the activity. The initial rate measurements were carried out in triplicate. One unit (U) was defined as the amount of protein that catalyzes the hydrolysis of 1 μmol *p*-nitrophenyl palmitate per minute in 0.1 M phosphate buffer (pH 7.0) at 25 °C. The specific activity of lipase in buffer was 27 U/mg.

### 2.5. Thermostability of lipase in aqueous solutions containing DES mixtures

The thermal stability of lipase in aqueous solutions containing DES mixtures was measured based on the residual activity of lipase after heat treatment. To 0.9 mL of 0.1 M phosphate buffer (pH 7.0) containing 10% DES mixture, 0.1 mL of lipase solution (10 μg protein/mL) was added. The mixture was incubated at 40 °C for a predetermined period of time (0, 0.25, 0.5, 1, 2, 4, 6, or 12 h), and then cooled in a cold-water bath for 2 min. Subsequently, the reaction was initiated by adding 8.5 mL of phosphate buffer (0.1 M, pH 7.0) containing 10% DES mixture and 0.5% Triton X-100, followed by 0.5 mL of substrate solution containing 10 mM *p*-nitrophenyl palmitate at 25 °C. Additionally, to investigate the effect of temperature on the stability of lipase in 10% [Ch]Cl:glycerol, residual activity of lipase was measured after 1 h incubation at various temperatures ranging from 30 °C to 55 °C. The residual activity of lipase was determined with the same procedure described above.

### 2.6. pH stability of lipase in aqueous solutions containing DES mixtures

The pH stability of lipase in aqueous solutions containing DES mixtures was measured based on the residual activity of lipase after treatment under acidic or basic conditions. To 0.9 mL of citric acid buffer (0.1 M, pH 4.0) or bicarbonate buffer (0.1 M, pH 9.0) containing 10% DES mixture, 0.1 mL of lipase solution (10 μg protein/mL) was added, and then the mixture was incubated at 25 °C. After incubation of lipase, the reaction was initiated by adding 8.5 mL of phosphate buffer (0.1 M, pH 7.0) containing 10% DES mixture and 0.5% Triton X-100, followed by 0.5 mL of substrate solution containing 10 mM *p*-nitrophenyl palmitate at 25 °C.

### 2.7. Storage stability of lipase in aqueous solutions containing DES mixtures

To 0.9 mL of 0.1 M phosphate buffer (pH 7.0) containing 10% DES mixture, 0.1 mL of lipase solution (10 μg protein/mL) was added, and the mixture was incubated at 25 °C for a predetermined period of time (0, 12, 24, or 48 h). Subsequently, the reaction was initiated by adding 8.5 mL of phosphate buffer (0.1 M, pH 7.0) containing 10% DES mixture and 0.5% Triton X-100, followed by 0.5 mL of substrate solution containing 10 mM *p*-nitrophenyl palmitate at 25 °C.

## 3. Results and discussion

### 3.1. Effect of DES mixtures as cosolvents on lipase activity

Fig. 1 shows the effect of DES mixtures on the hydrolytic activity of lipase when used as cosolvents in aqueous solution. Nine kinds of DES mixtures enhanced the activity of lipase in aqueous DES mixture solutions. The activity of lipase was enhanced up to 55% upon adding 10% [Ch]Cl:urea:glycerol to phosphate buffer (pH 7.0). On the other hand, the activity of lipase could not be enhanced by using only glycerol or ethylene glycol. In addition, to compare the effects of DES mixtures on lipase activity with the effects of ILs, [Bmim][$BF_4$], which has been used as a cosolvent for biocatalysis in aqueous solutions, was also tested [26]. However, this IL could not enhance the activity of lipase. These results indicate that the DES mixtures can be used as lipase activators.

When urea-containing DES mixtures other than [Ch]Cl:urea:formamide were used, the lipase activity increased from 16% to 55%. Interestingly, when urea and glycerol were used as HBD partners for DESs, a synergistic effect to increase lipase activity was found. The relative activities of lipase in [Ch]Cl:urea:glycerol, [Ch]Cl:urea, and [Ch]Cl:glycerol aqueous solutions were 155%, 122%, and 103%, respectively. The effect of the molar ratio of urea and glycerol in DES mixtures on the lipase activity was also studied (Fig. 2). The highest activity of lipase was obtained in the aqueous solution containing [Ch]Cl:urea:glycerol with the molar ratio of 1:1:1. On the other hand, when formamide-containing DES mixtures other than [Ch]Cl:ethylene glycol:formamide were used, the lipase activity decreased. Generally, urea, thiourea, and glycerol as HBD partners for DESs showed beneficial effects on lipase activity, while formamide decreased lipase activity. Durand et al. also reported the urea and glycerol are more efficient HBD partners than acetamide and ethylene glycol for lipase-catalyzed reaction in anhydrous DESs [9]. It means that the urea and glycerol are good HBD partners for DESs to enhance the activity of lipase when DESs are used as anhydrous reaction media or cosolvents for lipase-catalyzed reactions.

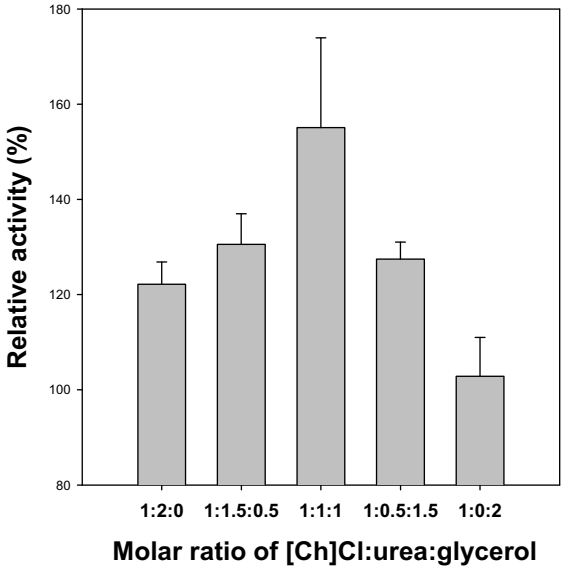



**Fig. 2.** Effect of DES molar ratio on the activity of lipase in aqueous DES solutions.

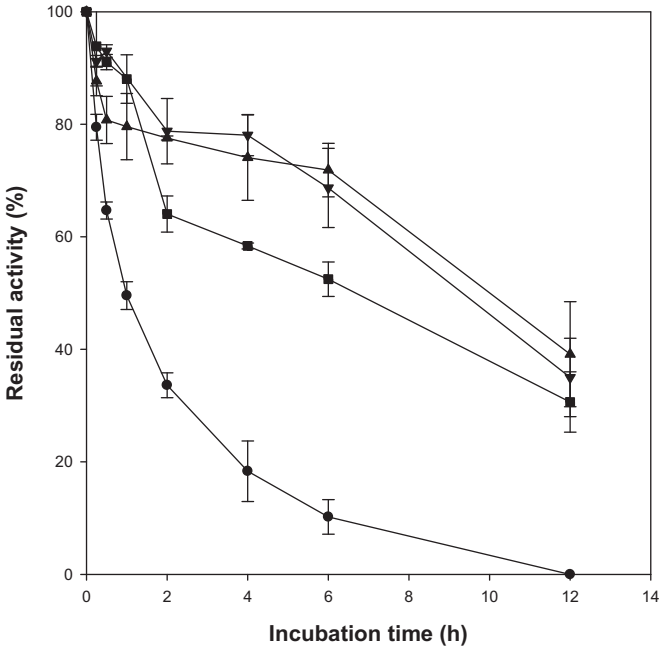



**Fig. 3.** Thermal stability of lipase in aqueous solutions containing DES mixtures at 40 °C (●: no additive, ■: 10% [Ch]Cl:ethylene glycol:formamide, ▼: 10% [Ch]Cl:glycerol, ▲: 10% [Ch]Cl:glycerol:ethylene glycol).

### 3.2. Effect of DES mixtures as cosolvents on thermal stability of lipase

Fig. 3 shows the thermal stability of lipase in aqueous solutions of DES mixtures at 40 °C. The residual activity of lipase in buffer after incubation for 1 h was approximately 50%, while the lipases in the aqueous solutions of DES mixtures retained over 80% of their initial activities. Lipase in buffer showed no activity after incubation for 12 h at 40 °C, while the residual activities of lipase in aqueous solutions of [Ch]Cl:ethylene glycol:formamide, [Ch]Cl:glycerol, and [Ch]Cl:glycerol:ethylene glycol were 30.6%, 35.0%, and 39.1%, respectively. These results clearly show the increased thermal stability of lipase in aqueous solutions of DES mixtures. The thermal stability of lipase in 10% [Ch]Cl:glycerol solution was also investi-

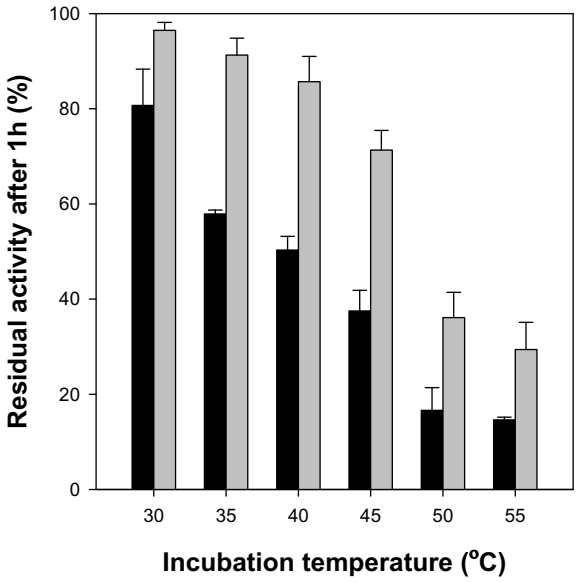



**Fig. 4.** Thermal stability of lipase in aqueous solutions containing 10% [Ch]Cl:glycerol at various temperatures (black bar: no additive, gray bar: 10% [Ch]Cl:glycerol).

gated at various temperatures (Fig. 4). At over 45 °C, the residual activity of lipase in 10% [Ch]Cl:glycerol after 1 h incubation was about 2-fold higher than that of lipase in buffer without additive.

Table 2shows the half-life times of lipase in aqueous solutions containing DES mixtures during incubation at 40 °C. All DES mixtures except [Ch]Cl:formamide:thiourea efficiently enhanced the thermal stability of lipase when they were used as cosolvents in buffer (pH 7.0). Interestingly, the half-life time of lipase in a 10% [Ch]Cl:glycerol solution was 9.2 times higher than that of lipase in buffer. Glycerol and ethylene glycol as HBD partners for DES mixtures could significantly enhance the thermal stability of lipase in aqueous DES solutions. The half-life time of lipase in aqueous solutions of DES mixtures containing ethylene glycol was greatly enhanced by 5.3–8.9 times compared to that of lipase in buffer. The half-life time of lipase in aqueous solutions of glycerol-containing DES mixtures was increased by 2.9–9.2 times, whereas formamide and thiourea as HBD partners for DES mixtures could not efficiently increase the thermal stability of lipase in aqueous solutions of DES mixtures. For example, [Ch]Cl:formamide:thiourea could not enhance the activity and thermal stability of lipase. Huang et al. also showed that glycerol and ethylene glycol as HBD partners for cholinium salt-based DESs are better than urea and acetamide to enhance the thermal stability of *P. expansum* lipase [18]. When only glycerol and ethylene glycol were used as cosolvents, they could also enhance the thermal stability of lipase. However, the lipase stabilizing effects of glycerol and ethylene glycol were lower than those of [Ch]Cl:glycerol and [Ch]Cl:ethylene glycol, respectively. When the changes in activity and stability of lipase in aqueous solutions of DES mixtures are considered, glycerol is a very useful HBD partner for DES mixtures for enhancing both the activity and stability of lipase. Whereas, formamide as a HBD partner for DES mixtures did not efficiently enhance the activity and stability of lipase.

### 3.3. Effect of DES mixtures as cosolvents on pH stability of lipase

Table 3 shows the pH stability of lipase in aqueous solutions of DES mixtures. No residual activity was observed when the lipase was incubated in a pH 4.0 citric acid buffer for 1 h. However, when the lipase was incubated in citric acid buffer containing 10% DES mixtures, the residual activity of lipase ranged from 3.7% to 57.1%. The lipase showed high acid stability in aqueous solutions of glycerol-containing DES mixtures, whereas low acid stability was observed in aqueous solutions of formamide-containing DES mixtures, although the residual activity of lipase was higher than that in buffer without DES mixtures.

The residual activity of lipase was only 3.8% after incubation in pH 9.0 bicarbonate buffer for 3 h, whereas when the lipase was incubated in bicarbonate buffer containing 10% DES mixtures, the residual activity of lipase ranged from 16.3% to 80.9%. The most efficient DES mixture at enhancing the base stability of lipase was [Ch]Cl:ethylene glycol:formamide. This DES mixture is very interesting, because formamide as a HBD partner for DES mixtures generally showed negative effects on the activity and thermal stability of lipase. However, the lipase activity at pH 7.0 and half-life time of lipase during incubation at 40 °C in an aqueous solution of [Ch]Cl:ethylene glycol:formamide were 117% and 610% higher than those in buffer without DES mixtures, respectively. The lipase showed high base stability in aqueous solutions of glycerol-containing DES mixtures, while the lipase showed low base stability in aqueous solutions of urea-containing DES mixtures.

### 3.4. Effect of DES mixtures as cosolvents on storage stability of lipase

Fig. 5 shows the storage stability of lipase in aqueous solutions of DES mixtures. The residual activity of lipase was 45% and 26% of the initial activity after incubation for 12 and 24 h in phosphate buffer, respectively. No lipase activity was observed after incubation for 48 h. Lipase stored in aqueous solutions of DES mixtures other than [Ch]Cl:formamide:thiourea showed higher residual activity than that stored in buffer without DESs after long incubation times. Glycerol-containing DES mixtures could efficiently maintain the activity of lipase for a long time. In particular, the highest residual activity of lipase after incubation for 48 h was obtained by using [Ch]Cl:glycerol:thiourea as a cosolvent, whereas urea- or formamide-containing DES mixtures could not efficiently maintain the activity of lipase. Among the tested DES mixtures, only [Ch]Cl:formamide:thiourea could not enhance the activity and stability of lipase.

### 3.5. Solvatochromic parameters of DES mixtures

The study of solvatochromic parameters of DES mixtures is very useful for helping to understand the effects of DES mixtures on lipase activity and stability, because solvatochromic parameters will explain the solvent properties of DES mixtures by comparing with those of other organic solvents and ILs. Table 4 shows the solvatochromic parameters of DES mixtures studied in this work.

The solvatochromic parameter $E_T^N$ is a normalized scale of $E_T(30)$, which is based on the intermolecular charge transfer absorption band of 2,6-diphenyl-4-(2,4,6-triphenyl-*N*-pyridino) phenolate (Reichardt's betaine dye) [27]. The solvatochromic properties of this dye are strongly affected by the hydrogen bond acidity of the solvent. The $E_T^N$ values of the DES mixtures studied in this work ranged from 0.814 to 0.857 and were much higher than those of protic solvents such as secondary amines (e.g., the value for *N*-methylacetamide are 0.657) and short-chain alcohols (the value for methanol is 0.762). The $E_T^N$ values of DES mixtures are very similar to that of glycerol (0.813) and slightly higher than ethylene glycol (0.792). Of the various reported ILs, hydrophilic ILs containing $N(CN)_2^-$ such as [Glymim]$N(CN)_2$ (0.831), [MeGlyMor]$N(CN)_2$ (0.836), and [MeGlyPyrr]$N(CN)_2$ (0.851) have similar $E_T^N$ values to those of DES mixtures [28]. Among the 13 studied DES mixtures, [Ch]Cl:glycerol:thiourea showed the highest $E_T^N$ value, while [Ch]Cl:ethylene glycol:formamide had the lowest value. The $E_T^N$ val-

**Table 3**
pH stability of lipase in aqueous solutions containing DES mixtures.

| DES mixture | Residual activity (%) after incubation for 1 h at pH 4.0 | Residual activity (%) after incubation for 3 h at pH 9.0 |
|---|---|---|
| Buffer (no additive) | 0.0 | 3.8 |
| [Ch]Cl:urea | 31.1 | 16.3 |
| [Ch]Cl:glycerol | 36.8 | 52.7 |
| [Ch]Cl:ethylene glycol | 18.8 | 38.4 |
| [Ch]Cl:urea:glycerol | 42.3 | 42.5 |
| [Ch]Cl:urea:ethylene glycol | 31.0 | 18.5 |
| [Ch]Cl:urea:formamide | 10.7 | 34.6 |
| [Ch]Cl:urea:thiourea | 38.6 | 24.8 |
| [Ch]Cl:glycerol:ethylene glycol | 35.6 | 47.1 |
| [Ch]Cl:glycerol:formamide | 22.8 | 77.6 |
| [Ch]Cl:glycerol:thiourea | 57.1 | 35.1 |
| [Ch]Cl:ethylene glycol:formamide | 31.0 | 80.9 |
| [Ch]Cl:ethylene glycol:thiourea | 28.3 | 34.0 |
| [Ch]Cl:formamide:thiourea | 3.7 | 29.7 |
| Glycerol | 29.5 | 37.0 |
| Ethylene glycol | 11.6 | 27.5 |

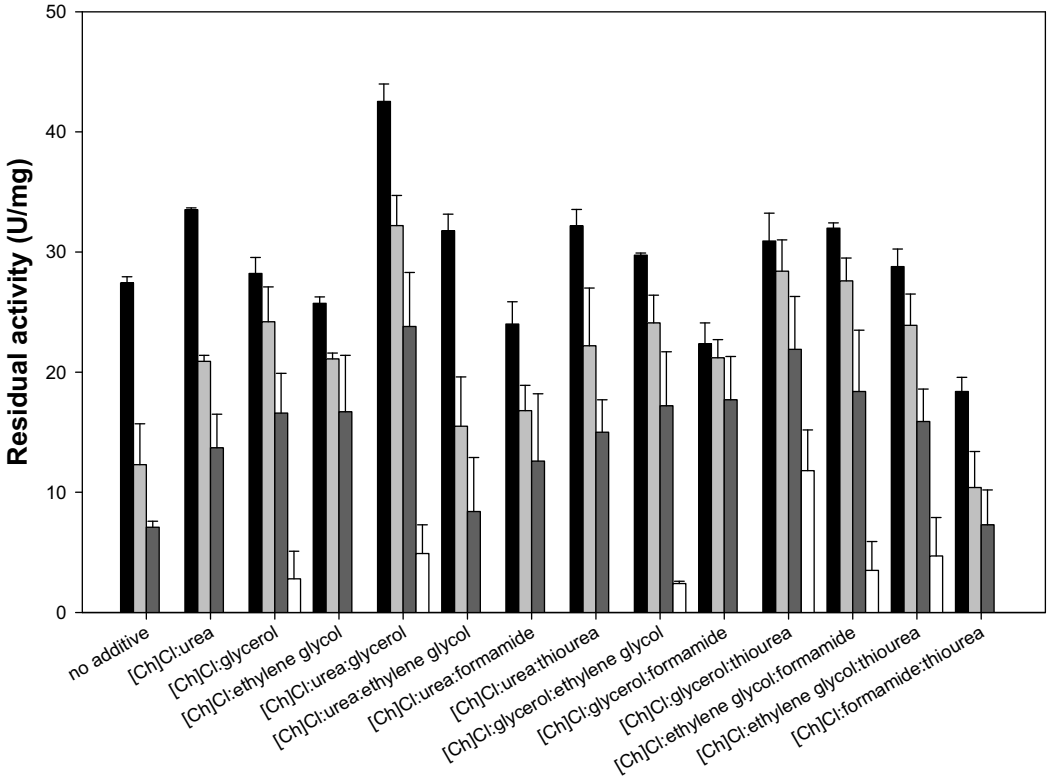



**Fig. 5.** Storage stability of lipase in aqueous solutions containing DES mixtures (black: 0 h, light gray: 12 h, dark gray: 24 h, white: 48 h).

ues of the DES mixtures increased with increasing $E_T^N$ value of the HBD partner for DESs in the following order: formamide ($E_T^N$ value is 0.775) < ethylene glycol (0.792) < urea < glycerol (0.813) < thiourea. DES mixtures containing thiourea generally had high $E_T^N$ values ranging from 0.833 to 0.857, while DES mixtures containing formamide had low $E_T^N$ values ranging from 0.814 to 0.833.

The solvatochromic parameter $\pi^*$ describes the combination of dipolarity and polarizability of the solvents. The $\pi^*$ values are determined by the bathochromic shifts of specified indicator compounds in polar solvents with cyclohexane as a reference solvent, and then normalized such that the value for dimethylsulfoxide is 1.00. The $\pi^*$ values of the DES mixtures studied in this work ranged from 1.112 to 1.245 and were higher than those of water ($\pi^*$ value is 1.09) and dimethylsulfoxide (1.00) were. The $\pi^*$ values of DES mixtures are very similar to those of glycerol (1.139) and 2-cyanopyridine (1.2). Of the reported ILs, morpholinium-based ILs containing $N(CN)_2^-$ such as [MeEtMor]$N(CN)_2$ (1.12) and [MeGlyMor]$N(CN)_2$ (1.20) have similar $\pi^*$ values to those of DES mixtures [28]. Among the 13 studied DES mixtures, [Ch]Cl:urea:thiourea showed the highest $\pi^*$ value, while [Ch]Cl:ethylene glycol had the lowest value. The $\pi^*$ values of the DES mixtures increased upon changing the HBD partner for DESs in the following order: ethylene glycol < formamide < glycerol < urea < thiourea. Thiourea had the greatest influence among HBD partners for DES mixtures in terms of increasing the $\pi^*$ values of the DES mixtures. The $\pi^*$ values of DES mixtures containing thiourea ranged from 1.207 to 1.245 and were higher than previously reported general ILs and organic solvents.

The α scale is a measure of solvent hydrogen bond acidity normalized to a value of 1.00 for methanol. The α values of the DES mixtures studied in this work ranged from 0.840 to 0.903 and were similar to those of short-chain alcohols (e.g., the val-

**Table 4**
Solvatochromic parameters of the DES mixtures.

| DES mixture | $E_T^N$ | $\pi^*$ | α | $\beta$ |
|---|---|---|---|---|
| [Ch]Cl:urea | 0.835 | 1.192 | 0.860 | 0.559 |
| [Ch]Cl:glycerol | 0.845 | 1.161 | 0.903 | 0.544 |
| [Ch]Cl:ethylene glycol | 0.822 | 1.112 | 0.891 | 0.640 |
| [Ch]Cl:urea:glycerol | 0.843 | 1.170 | 0.893 | 0.583 |
| [Ch]Cl:urea:ethylene glycol | 0.827 | 1.165 | 0.862 | 0.560 |
| [Ch]Cl:urea:formamide | 0.817 | 1.159 | 0.846 | 0.593 |
| [Ch]Cl:urea:thiourea | 0.844 | 1.245 | 0.840 | 0.486 |
| [Ch]Cl:glycerol:ethylene glycol | 0.830 | 1.127 | 0.897 | 0.625 |
| [Ch]Cl:glycerol:formamide | 0.829 | 1.131 | 0.891 | 0.598 |
| [Ch]Cl:glycerol:thiourea | 0.857 | 1.217 | 0.889 | 0.539 |
| [Ch]Cl:ethylene glycol:formamide | 0.814 | 1.113 | 0.872 | 0.657 |
| [Ch]Cl:ethylene glycol:thiourea | 0.842 | 1.207 | 0.864 | 0.601 |
| [Ch]Cl:formamide:thiourea | 0.833 | 1.209 | 0.843 | 0.549 |
| Glycerol | 0.813 | 1.139 | 0.851 | 0.597 |
| Ethylene glycol | 0.792 | 1.014 | 0.898 | 0.620 |
| [Bmim][$BF_4$] | 0.670 | 1.046 | 0.626 | 0.446 |

ues for ethanol and 1-propanol are 0.86 and 0.84, respectively), but were much higher than those of many organic solvents. Of the reported ILs, those containing $N(CN)_2^-$ such as [Glymmim]$N(CN)_2$ (0.84), [MeGlyPip]$N(CN)_2$ (0.86), and [MeGlyPyrr]$N(CN)_2$ (0.90) have similar α values to those of DES mixtures [28]. Among the 13 studied DES mixtures, [Ch]Cl:glycerol showed the highest α value, while [Ch]Cl:urea:thiourea had the lowest value. The α values of DES mixtures increased upon changing the HBD partner for DESs in the following order: thiourea < formamide < urea < ethylene glycol < glycerol. DES mixtures containing glycerol generally had high α values ranging from 0.889 to 0.903, while DES mixtures containing thiourea had low α values ranging from 0.840 to 0.889.

The β scale is a measure of the solvent hydrogen bond basicity normalized to a value of 1.00 for hexamethylphosphoramide. It appeared that the hydrogen bond accepting abilities of the DES mixtures studied here covered a wide range, from those of acetone (0.48) and water (0.49) through to those of ethyl acetate (0.63) and glycerol (0.67). The low β values of the DES mixtures were similar to those of ILs containing $N(CN)_2^-$ such as [MeGlyPyrr]$N(CN)_2$ (0.46) and [Glymim]$N(CN)_2$ (0.47) [28]. The β values of DES mixtures increased upon changing the HBD partner for DESs in the following order: thiourea < glycerol < urea < ethylene glycol < formamide. DES mixtures containing formamide generally had high β values ranging from 0.549 to 0.657, while DES mixtures containing thiourea had low β values ranging from 0.486 to 0.601.

### 3.6. Influence of solvatochromic parameters of DES on the activity or stability of lipase

To understand the effect of DES mixtures on the activity and stability of lipase in aqueous DES solutions, the correlations between solvatochromic parameters of DES mixtures and the activity and stability of lipase were analyzed.

When the solvatochromic parameters of DES mixtures were correlated with the lipase activity in aqueous solutions of DES mixtures, unfortunately, a good relationship was not found with Kamlet-Taft parameters. In general, the activity of lipase increased with increasing $E_T^N$ value of the DES mixtures, although the correlation coefficient ($r$) of 0.316 ($n$ = 13) was low. However, better correlation with higher $r$ value of 0.546 ($n$ = 15) could be obtained for the data set including DES mixtures, glycerol, and ethylene glycol. Therefore, enhanced activity of lipase in aqueous DES solutions can be partially understood by the unique high $E_T^N$ values of DES mixtures. Park and Kazlauskas also showed similar correlation between $E_T^N$ value of IL and activity of *Pseudomonas cepacia* lipase [23]. They reported the activity of lipase in anhydrous ILs increased with increasing the $E_T^N$ values of ILs.

When the solvatochromic parameters of DES mixtures were correlated with the thermal stability of lipase in aqueous solutions of DES mixtures, it was found that the α value was closely associated to the thermal stability of lipase, with the highest correlation coefficient ($r$ = 0.701) among the four solvatochromic parameters ($E_T^N$, $\pi^*$, α, β). Generally, the thermal stability of lipase increased with increasing α value of the DES mixtures. When the solvatochromic parameters of the DES mixtures were correlated with the storage stability of lipase in aqueous solutions of DES mixtures, it was also found that the α value was closely related to the storage stability of lipase, with the highest correlation coefficient among the four solvatochromic parameters. The correlation coefficient values were 0.677 and 0.648, respectively, for the correlation between the residual activity after incubation for 12 and 24 h and the α value. Generally, the storage stability of lipase increased with increasing α value of the DES mixtures. Therefore, we propose that the stabilizing effect of DES mixtures on the lipase may be caused by strong hydrogen bond donating power of DES mixtures. Increased interactions by the hydrogen bond formation between lipase and DES mixtures may increase the thermal stability and storage stability of lipase in aqueous DES solutions.

When the solvatochromic parameters of DES mixtures were correlated with the acid stability of lipase in aqueous solutions of DES mixtures, it was found that the $E_T^N$ value was the most well correlated parameter among the four solvatochromic parameters, with a correlation coefficient of 0.666. Generally, the acid stability of lipase increased with increasing $E_T^N$ value of the DES mixtures. When the solvatochromic parameters of DES mixtures were correlated with the base stability of lipase in aqueous solutions of DES mixtures, it was found that the $\pi^*$ value was the most well correlated parameter of the four solvatochromic parameters, with a correlation coefficient of 0.642. Generally, the base stability of lipase increased with increasing $\pi^*$ value of the DES mixtures. At acidic or basic conditions, the determining factor among four solvatochromic parameters for pH stability of lipase was different from the most influencing factor for thermal stability and storage stability. It may be caused by the change of ionization state in the surface of lipase at acidic or basic conditions.

## 4. Conclusions

DES mixtures could efficiently enhance the activity, thermal stability, pH stability, and storage stability of lipase when used as cosolvents in aqueous solution. In general, glycerol-containing DES mixtures were very useful in enhancing the activity and stability of lipase, while formamide-containing DES mixtures could not efficiently enhance the activity and stability of lipase. Urea-containing DES mixtures could enhance the activity of lipase, but could not efficiently increase the stability of lipase. When the solvatochromic parameters of DES mixtures were correlated with the stability of lipase in aqueous solutions of DES mixtures, it was found that the thermal stability and storage stability of lipase were associated with the α value of the DES mixtures, while the pH stability of lipase under acidic and basic conditions was related to the $E_T^N$ and $\pi^*$ values of the DES mixtures, respectively.

In this work, it was clearly shown that DES mixtures as cosolvents effectively enhance the activity and stability of lipase. DES mixtures have many potential applications as protein activators or stabilizers because DES mixtures are cheap, environmentally friendly, biocompatible, and biodegradable. In addition, the properties of DES mixtures that influence the activity and stability of lipase could be partially understood through the correlation with solvatochromic parameters. Therefore, an optimal DES mixture as a protein stabilizer can be easily prepared via the combination of various HBDs at different mixing ratios.

## Acknowledgements

This paper was written as part of Konkuk University's research support program for its faculty on sabbatical leave in 2015. This work was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2015R1D1A1A01060206). This work was also supported by the Korea CCS R&D Center (KCRC) grant (2014M1A8A1049296) and the Rural Development Administration through the Cooperative Research Program for Agriculture Science & Technology Development (010205022014).

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